

12.289

AI25BTECH11003 - Bhavesh Gaikwad

October 11, 2025

Question

The approximate eigenvalue of the matrix

$$\mathbf{A} = \begin{pmatrix} 15 & 4 & 3 \\ 10 & 12 & 6 \\ 20 & 4 & 2 \end{pmatrix}$$

obtained after two iterations of Power method, with the initial vector $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$ is

(MA 2012)

- (a) 7.768 (b) 9.468 (c) 10.548 (d) 24.145

Theoretical Solution

Given:

$$\mathbf{A} = \begin{pmatrix} 15 & 4 & 3 \\ 10 & 12 & 6 \\ 20 & 4 & 2 \end{pmatrix}$$

Let the initial vector be $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$\mathbf{x}_1 = \mathbf{A}\mathbf{x}_0 = \begin{pmatrix} 22 \\ 28 \\ 26 \end{pmatrix} \quad (1)$$

Let Normalization vector: $\mathbf{y}_i = \frac{1}{x_{max}} \mathbf{x}_i$
where, x_{max} is the largest element in $\mathbf{x}_i$

$$\mathbf{y}_1 = \frac{1}{28} \mathbf{x}_1 \quad (2)$$

Theoretical Solution

$$\mathbf{x}_2 = \mathbf{A}\mathbf{y}_1 = \frac{1}{7} \begin{pmatrix} 260 \\ 356 \\ 302 \end{pmatrix} \quad (3)$$

$$\mathbf{y}_2 = \frac{1}{356} \mathbf{x}_2 = \frac{1}{1246} \begin{pmatrix} 130 \\ 178 \\ 151 \end{pmatrix} \quad (4)$$

Let λ be the Dominant eigenvalue.

By Rayleigh-Quotient,

$$\lambda = \frac{\mathbf{y}_2^\top \mathbf{A} \mathbf{y}_2}{\mathbf{y}_2^\top \mathbf{y}_2} \quad (5)$$

$$\lambda = 24.1453$$

Thus Option-D is Correct